

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think>

- CD 2 1 is a receptor for the Epstein-B arr virus (EB V).
- EB V binds to CD 2 1 on B cells during infection .
- Without CD 2 1 , B cells will not be able to interact with EB V effectively , leading to a decreased path ogenesis of EB V .
- Me as les , HIV , and human pap ill om avirus do not directly interact with CD 2 1 in the way EB V does .
- The genetic modification of B cells to not express CD 2 1 will most likely affect the path ogenesis of EB V the most .
- This is because EB V specifically uses CD 2 1 as a receptor for infection .
- Therefore , the path ogenesis of EB V would be significantly affected .
- The other viruses listed do not have this specific interaction with CD 2 1 .
- Answer A (Ep stein -B arr virus) is the correct answer .
- The other viruses (me as les , HIV , and human pap ill om avirus) do not interact with CD 2 1 in the same way .
- The modification of CD 2 1 would not significantly affect the path ogenesis of these other viruses .
- The other answer choices (me as les virus , HIV , and human pap ill om avirus) are less likely to be affected by this genetic modification .
- The question is asking for the organism whose path ogenesis would be most likely affected by this genetic modification .
- The most affected organism is EB V .
- The answer is A (Ep stein -B arr virus) .

</think>< answer>A . Epstein-B arr virus (EB V)</answer>